Olivier Côté ACAS

PhD Candidate in Actuarial Science, Université Laval

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Languages: Français (native), English (C1), Svenska (A1) | Programming: R, Python, LaTeX

Education

PhD in Actuarial Science (GPA 4.17/4.33) 2023–2027
Université Laval
Fairness in insurance pricing, supervised by Prof. Marie-Pier Côté (Université Laval) and Prof. Arthur Charpentier (UQÀM). Defence expected in summer 2027.

Master's degree in Actuarial Science, research-based (GPA 4.24/4.33) 2021–2023
Université Laval
Supervised by Prof. Marie-Pier Côté and Prof. Arthur Charpentier.

Bachelor's degree in Actuarial Science (GPA 4.17/4.33) 2018–2021
Université Laval

Research visits 2023–2026
Stockholm University, University of Toronto, Bayes Business School

- Department of Mathematics, Stockholm University (two weeks, January 2026): invited seminar and research discussions with Prof. Mathias Millberg Lindholm.
- Department of Statistics, University of Toronto (2025): invited seminar and research discussions with Prof. Christopher Blier-Wong.
- Bayes Business School, London (one week, 2023): research discussions with Prof. Andreas Tsanakas.

Research papers

Côté, O., Côté, M.-P., and Charpentier, A. (2027). A statistical perspective on fairness in insurance pricing. *Annual Review of Statistics and Its Application*. Invited review, accepted June 2026. In press

Charpentier, A., **Côté, O.**, and Côté, M.-P. (2026b). Governing fairness in insurance pricing: The trilemma of actuarial adequacy, solidarity, and causal legitimacy. *Journal of Business Ethics*. Published online September 2026, doi:10.1007/s10551-026-06479-2

Côté, M.-P., **Côté, O.**, and Charpentier, A. (2026). Recoverability of market-wide fair insurance premiums under selection bias. *Insurance: Mathematics and Economics*, 129:103268. doi:10.1016/j.insmatheco.2026.103268

Côté, O., Côté, M.-P., and Charpentier, A. (2026). A scalable toolbox for exposing indirect discrimination in insurance rates. *CAS Research Paper Series on Bias and Insurance*. Research paper, casact.org

- Winner of the American Academy of Actuaries Award for Research 2026
- Selected by the National Association of Insurance Commissioners (NAIC) for its 2026 webinar series for U.S. insurance regulators and their actuarial staff

Charpentier, A., Fernandes Machado, A., **Côté, O.**, and Hu, F. (2026a). Federated measurement of demographic disparities from quantile sketches. *arXiv:2602.18870*. Preprint

Côté, O., Côté, M.-P., and Charpentier, A. (2025). A fair price to pay: Exploiting causal graphs for fairness in insurance. *Journal of Risk and Insurance*, 92(1):33–75. doi:10.1111/jori.12503

- Finalist for the American Academy of Actuaries Award for Research 2025

Work in progress

Côté, O., Charpentier, A., Côté, M.-P., and Il Idrissi, M. A definition of confounding bias for arbitrary statistical functionals. Working paper, in preparation.

Côté, O., Côté, M.-P., Charpentier, A., and Il Idrissi, M. Profiling actuarial discrimination via causal decomposition. Working paper, presented at the 29th International Congress on Insurance: Mathematics and Economics, Seoul.

Diversity, equity and inclusion education

Get Comfortable Being Uncomfortable: Dialogue About Favoritism and Fairness 2025
Canadian Statistical Sciences Institute

Let's talk about race and collective responsibility 2024
Canadian Statistical Sciences Institute

Acting as a sentinel for the prevention of suicide 2023
Centre de prévention du suicide du Québec workshop

Addressing Conflicts Related to Bias, Privilege, and Identity in the STEM Fields 2022
CANSSI and Academic Impressions workshop

Work experience

Course lecturer, two mandatory undergraduate courses (3 credits each) 2022–2023
Université Laval

- Linear Models in Actuarial Science (fall 2022, 88 students)
- Statistical Learning in Actuarial Science (winter 2023, 54 students)

Awards: Faculty of Science and Engineering Teaching Star, one per course (fall 2022 and winter 2023).

American football coach 2019–2025
Falcons of Cégep de Lévis
American football coach for different teams for more than ten years (2012–2025), including seven years in collegiate football at the Cégep de Lévis (2019–2025):

- Tight ends and Fullbacks coach (2019–2025), responsible for the development of 10 athletes
- Supervisor for a bachelor's degree intern in sports intervention (2023)
- Volunteer mathematics tutor for student-athletes

Awards: RSEQ Coaching Staff of the Year 2023 and 2024, Sportsmanship Award 2022 (D1) and 2023 (D2).

Teaching assistant 2019–2024
Université Laval
Statistical Learning in Actuarial Science (5 times), Probabilistic Analysis of Actuarial Risks (3 times), Linear Models in Actuarial Science (once), internship evaluation (once).

Actuarial intern 2019–2021
Desjardins General Insurance Group
Four internships: property ratemaking (2019), business analysis (2020 and 2021) and research (2021).

Supervision and mentoring

Mentor of a master's student, University of Texas 2025–2026
Hickman Scholars Program network
Advised an essay on an insurance regulatory issue and connected the student with a Texas regulator.

Co-supervisor of a master's student, Université Laval 2023
With Prof. Marie-Pier Côté
One-year project on fairness in insurance. The student now works as an actuary.

Supervisor of a PhD student, Université Laval Ongoing
Mitacs Accelerate partnership
Helped design the thesis project of the student who continues the industrial partnership.

Engagement with practice and policy

Online course on fair insurance pricing, English and French versions 2025–2026
Casualty Actuarial Society (in post-production)
Commissioned by the CAS. Project lead and script writer, coordinating four parties: a research assistant, the CAS task force on biases in insurance pricing, the CAS educational development team and a production company. Two hours of two-minute videos, interactive activities and quizzes, co-presented with M.-P. Côté and A. Charpentier.

Regulators, insurers and public bodies 2023–2026
Canada, United States, Switzerland

- Met insurance regulators in person in Canada, the United States and Switzerland.
- Provided input, in person, to the Swiss Federal Statistical Office and the ETH Zurich AI Policy Lab on a governmental framework for fair predictive models in employment.
- Presented at the internal annual actuarial conferences of three insurers.
- Three companies contacted me for technical details on implementing the toolbox with their own data.

Refereeing

European Actuarial Journal 2026–present
Annals of Actuarial Science 2026–present
Journal of Risk and Insurance 2024–present

Scholarships and awards

Award for Research 2026
American Academy of Actuaries
Winner (honorarium US$7 500, ≈ 75 000 SEK) for *A scalable toolbox for exposing indirect discrimination in insurance rates*. Annual international competition for papers by early-career scholars.

Travel Grant 2026
NSERC General Fund
Travel grant to support conference participation and dissemination of research ($1 300 CAD, ≈ 9 100 SEK).

Actuarial Research Conference 2026 Travel Grant 2026
American Academy of Actuaries
Travel grant to support conference participation and dissemination of research (US$1 500, ≈ 15 000 SEK).

Student travel grant 2026
Big Data Research Center
Travel grant to support conference participation and dissemination of research ($2 000 CAD, ≈ 14 000 SEK).

Canada Graduate Research Scholarship – Doctoral 2025
Natural Sciences and Engineering Research Council of Canada
National competition, $120 000 CAD (≈ 841 000 SEK) over three years, held from May 2025.

Hickman Scholars Program 2024, 2025, 2026
Society of Actuaries
International competition, US$20 000 (≈ 199 000 SEK) per year for three years (US$60 000 in total, ≈ 598 000 SEK). Every annual renewal is competitive.

Doctoral Training Scholarships 2024
Fonds de recherche du Québec – Nature et technologies
Provincial competition, $100 000 CAD (≈ 701 000 SEK) offered over four years, held from May 2024 to May 2025, when the NSERC doctoral scholarship began.

Mitacs Accelerate 2024
Mitacs
Research partnership with a large Canadian P&C insurer, $84 000 CAD (≈ 589 000 SEK) over three years (2024–2027), in which I am the lead student researcher. Selected after a scientific peer review and an internal Mitacs review.

Doctorate attraction grant Florent Toureille 2023
Chaire d'actuariat de l'Université Laval
Institutional competition based on leadership and excellence, $30 000 CAD (≈ 210 000 SEK) over three years.

Leadership and Engagement Scholarship 2023
Université Laval
Institutional competition in the sports sector, $30 000 CAD (≈ 210 000 SEK) over three years.

Charles Brindamour Graduate Scholarship for Excellence 2023
Université Laval, funded by Charles Brindamour (CEO of Intact Financial Corporation)
Institutional competition based on excellence, one award a year, $10 000 CAD (≈ 70 000 SEK).

François-Laviolette Scholarship 2022
Université Laval
Institutional competition for responsible artificial intelligence, $7 500 CAD (≈ 53 000 SEK).

Grant of attraction to the Master's 2021
Chaire d'actuariat de l'Université Laval
Institutional competition recognizing research potential, $15 000 CAD (≈ 105 000 SEK).

CAS Trust Scholarship 2021
Casualty Actuarial Society
International competition, US$2 500 (≈ 25 000 SEK).

Alexander Graham Bell Canada Graduate Scholarship – Master's 2021
Natural Sciences and Engineering Research Council of Canada
National competition, $17 500 CAD (≈ 123 000 SEK).

Lieutenant Governor's Youth Medal 2018
Cégep de Lévis
Awarded for outstanding involvement and academic results, on nomination by the teaching staff.

Professional designations

Associate of the Casualty Actuarial Society (ACAS) 2025
Casualty Actuarial Society

Professionalism Workshop 2025
Canadian Institute of Actuaries

MAS-I, MAS-II, Exam 5, Exam 6C, Online Courses 1 and 2 2020–2025
Casualty Actuarial Society

Exam P, FM and IFM 2019–2020
Society of Actuaries and Canadian Institute of Actuaries (credits)

Academic communications

Operationalizing Fairness in Actuarial Pricing: From Principle to Practice 2026
Casualty Actuarial Society webinar
Sole presenter for a 90-minute webinar to 950 registrants. *Audience: practitioners.*

The Trilemma of Fairness in P&C Pricing 2026
61st Actuarial Research Conference (AAA invited session)
Invited presentation. *Audience: decision-makers.*

Profiling Actuarial Discrimination via Causal Decomposition 2026
29th International Congress on Insurance: Mathematics and Economics, Sungkyunkwan University, Seoul (South Korea)
Academic presentation. *Audience: academics.*

Le trilemme de l'équité en tarification IARD 2026
Association des actuaires en IARD du Québec (AAIARD)
Presentation to P&C actuaries. *Audience: practitioners.*

Profiling Actuarial Discrimination via Causal Decomposition 2026
STATQAM seminar, Université du Québec à Montréal
Invited seminar covering causal decomposition and confounding bias. *Audience: academics.*

Profiling Actuarial Discrimination via Causal Decomposition 2026
Centre de recherche en données massives, Université Laval
Research talk. *Audience: academics.*

Profiling Actuarial Discrimination via causal decomposition 2026
Actuarial and Financial Mathematics Conference, Brussels (Belgium)
Poster presenter. *Audience: academics.*

Bias in Assessing Financial Risk: Insights from the 2025 Research Award 2026
American Academy of Actuaries
Invited panelist as finalist for the 2025 Award for Research on Bias in Assessing Financial Risk. Webinar to 445 actuaries and practitioners. *Audience: practitioners.*

A Causal Perspective on Fairness in Insurance Pricing 2026
Department of Mathematics, Stockholm University
Invited seminar talk during research stay. *Audience: academics.*

A Scalable Toolbox for Exposing Indirect Discrimination In Insurance Rates 2026
ASTIN Bulletin Conference, Zurich (Switzerland)
Speaker at one of the fairness sessions of the first ASTIN Bulletin Conference. *Audience: academics.*

Operationalizing Fairness in Actuarial Pricing: From Principle to Practice 2025
Department of Statistics, University of Toronto
Invited seminar during research visit. *Audience: academics.*

Operationalizing Fairness in Actuarial Pricing: From Principle to Practice 2025
Casualty Actuarial Society Annual Meeting, Austin (United States) & Canada Connect Day, Toronto (Canada)
Invited presenter to over 150 (Austin) and over 100 (Toronto) actuaries and policymakers. *Audience: practitioners.*

Two invited presentations at the 60th Actuarial Research Conference 2025
American Academy of Actuaries panel & Casualty Actuarial Society session, Toronto (Canada)
Guest panelist and presenter (with Marie-Pier and Arthur) to audiences from actuarial organizations, academia, industry, and policymakers. *Audience: decision-makers.*

A Scalable Toolbox for Exposing Indirect Discrimination in Insurance Rates 2025
Perspectives on Actuarial Risks in Talks of Young Researchers (PARTY), Liverpool (U.K.)
Presenter to 60 doctoral students from 29 countries. *Audience: academics.*

Une boîte à outils pour dévoiler la discrimination indirecte dans les tarifs d'assurance 2025
Chaire ACTIONS & Milliman R&D, Paris (France)
Guest co-presenter (with Arthur) to audiences from academia and industry (80 participants at the Chaire ACTIONS). *Audience: academics and practitioners.*

Perspective causale sur les discriminations dans les modèles prédictifs en assurance 2024
Institut intelligence et données
Guest presenter to over 100 people from academia. *Audience: academics.*

A Fair price to pay: exploiting causal graphs for fairness in insurance 2024
27th International Congress on Insurance: Mathematics and Economics
Academic presentation in Chicago, IL, United States. *Audience: academics.*

A Fair price to pay: exploiting causal graphs for fairness in insurance 2024
Workshop on fairness and discrimination in insurance
Guest presenter to over 60 people. *Audience: academics, practitioners and regulators.*

Perspective causale sur la discrimination indirecte en assurance générale 2024
Autorité des marchés financiers
Presentation to decision-makers at Québec's financial authority, followed by interviews with ten regulators on their needs. *Audience: decision-makers.*

Équité et discrimination en assurance générales 2023
Intact Financial Corporation
Presentation to CEO, CLO and CDO of Intact Financial Corporation. *Audience: decision-makers.*

Causal inference and Fairness in Insurance Pricing 2023
Insurance Data Science Conference
Presenter for Insurance Data Science Conference in London (UK). *Audience: academics.*

Public outreach

L'avenir à travers une boule de cristal mathématique : Le travail d'un actuaire 2023–2026
Cégep de Lévis
Guest presenter for CEGEP students. *Audience: students.*

Pour une intelligence artificielle équitable en assurance 2024
Université Laval News
I collaborated for a scientific communication article. *Audience: general public.*

Perspective causale sur la discrimination indirecte en assurance générale 2024
Institut Intelligence et données
4th research competition on data intelligence. ≈ 50 graduate students. *Audience: general public.*

Perspective causale sur la discrimination indirecte en assurance générale 2024
Institut de valorisation des données
I presented my research during IVADO Digital Future Day 2024. Over 250 people. *Audience: general public.*

L'IA pour calculer le prix de votre assurance auto: poétique ou pas éthique? 2023
Big Data Research Center
Presentation during the event *lightning talks 180*. *Audience: general public.*

Awards: Public's choice award

Artificial intelligence to prevent unfair discrimination in auto insurance 2023
IVADO
I authored an 800-word scientific communication article. *Audience: general public.*

Organizing

Workshop on fairness and discrimination in insurance 2026
Université Laval, 6–7 October 2026, 90 participants
Member of the organizing committee, host of the second day, panel moderator and presenter.

Quantact Student Day 2025
Province-wide event, 3 October 2025
Organizer: recruited a five-member committee, coordinated seven presenters and hosted sixty students.

Committees

Institut Intelligence et Données, Université Laval 2025–present
Voting member of the board

Actuarial Science Graduate Program Committee of Université Laval 2022–present
Voting member of the Committee

Big Data Research Center of Université Laval 2022–2025
Student voting member of the Committee

Actuarial Science Student Investment Fund of Université Laval 2021–2024
Voting member of the committee

Research affiliations

Institut Intelligence et Données, Big Data Research Center 2021–present
Université Laval

Quantact research laboratory, Centre de recherches mathématiques 2021–present
Montréal

Other

Roundtable on raising awareness about youth bullying 2025
Mammouth hebdo - Téléquébec

Actuarial School representative at the open house 2022–2023
Faculty of Science and Engineering

- Winter 2023 (≈ 30 visits from students accompanied by their parents)
- Fall 2022 (≈ 50 visits from students accompanied by their parents)